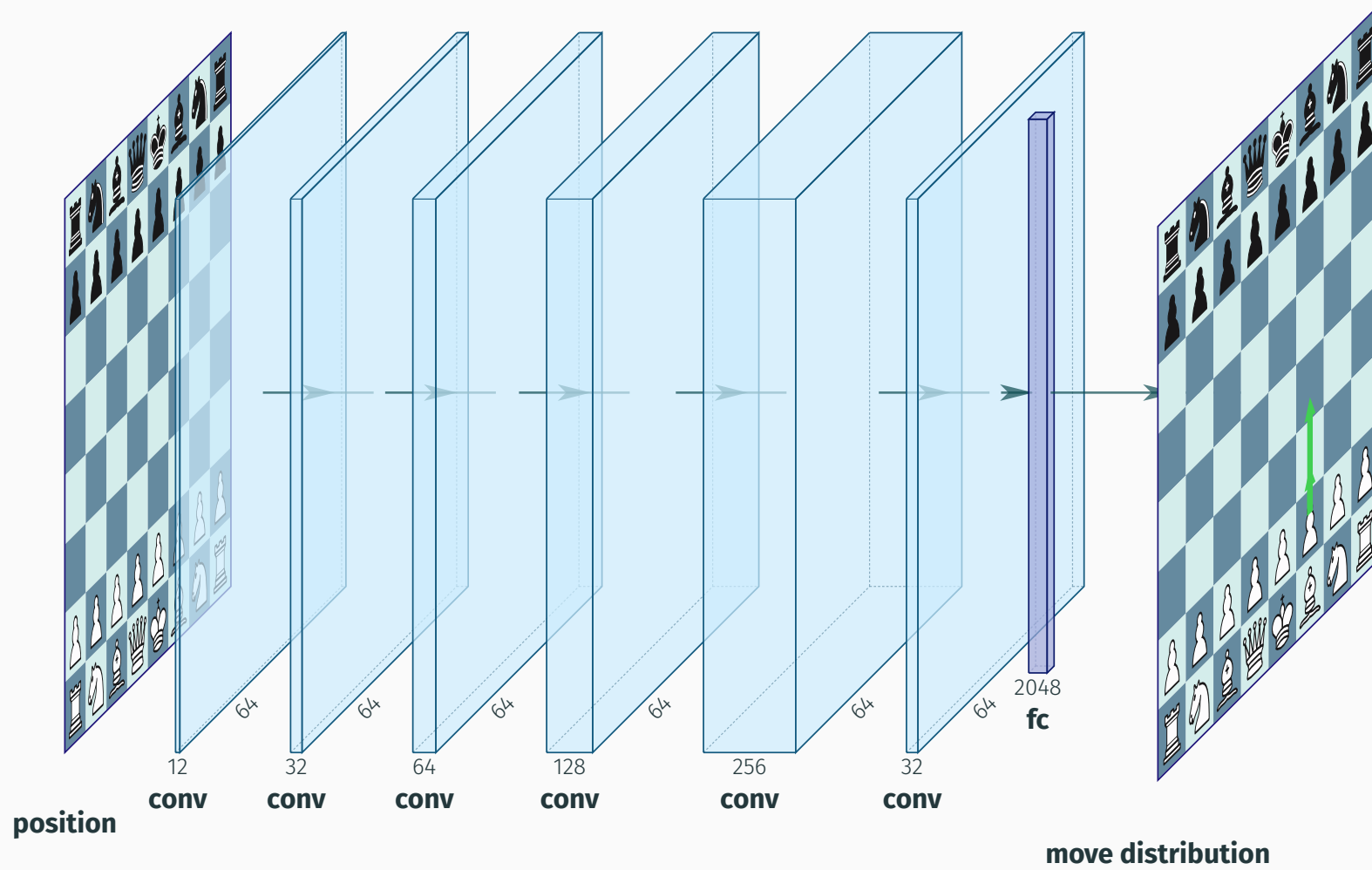


Exploring Behavior Cloning and Reward Shaping

Jakob Lambert-Hartmann

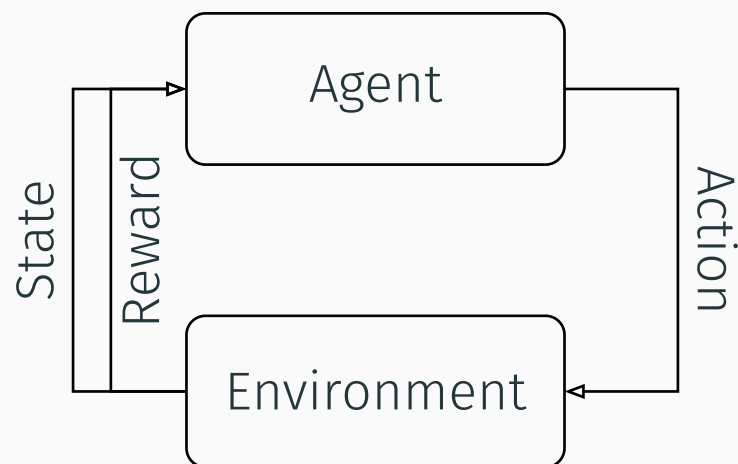
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Introduction



Reinforcement Learning

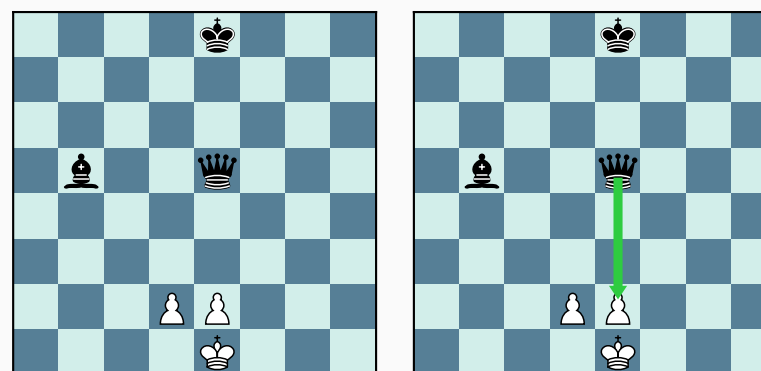
Reinforcement Learning



Problems

- Sparse rewards → poor performance
- Reward Shaping → difficult:
 - Missaligned rewards
 - Reward hacking
 - Prioritizing wrong rewards

Behavior Cloning



- Supervised learning
- Agent learns expert demonstrations
- e.g. puzzles, grandmaster games

Problems

- Demonstration sample bias → poor performance

Combine both approaches

1. Pretrain: Behavior Cloning
2. Finetune: Reward Based Reinforcement Learning

Can both approaches stabilize each other?

- Behavior Cloning: Stabilize RL with sparse rewards
- Reinforcement Learning: Compensates for data bias

Lets find out!

- Train different models:
 - Architecture: Fully connected, cnn, residual block
 - Pretraining Data: Heavily biased → less biased
 - Rewards Sets: Sparse → Dense

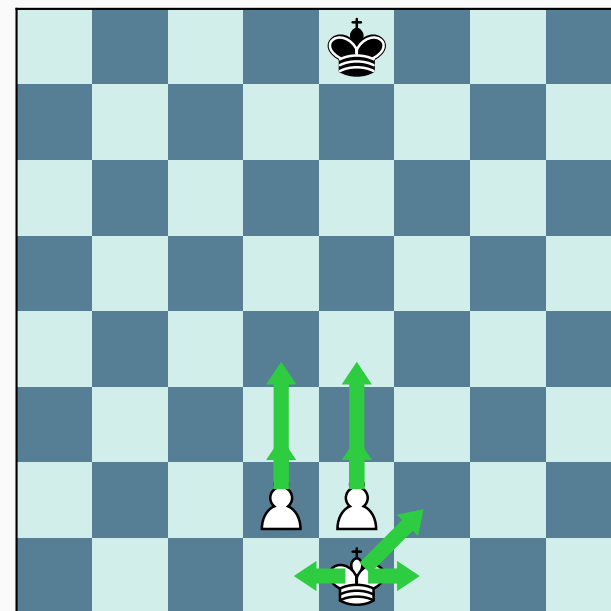
Methods

Input

- tensor shape = $(8, 8, 12)$:
 - 8×8 squares
 - 12 pieces: {White, Black} $\times$ {Pawn, Rook, Knight, Bishop, Queen, King}

Output

- Move distribution: from square $\rightarrow$ to square
- **But:** Not all moves are legal $\rightarrow$ mask illegal moves



Architecture

FCNN

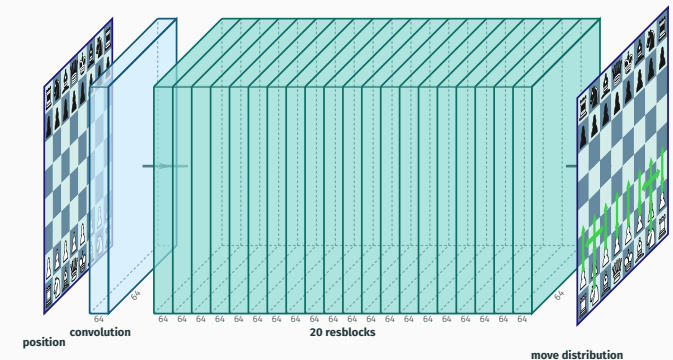
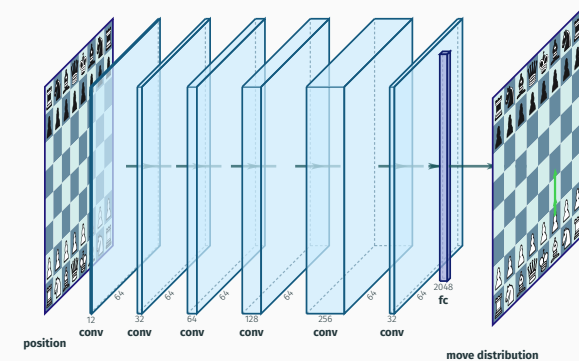
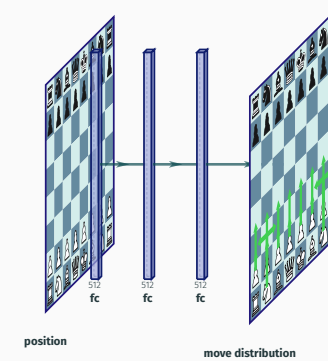
- 3 hidden layers: 512 neurons

CNN

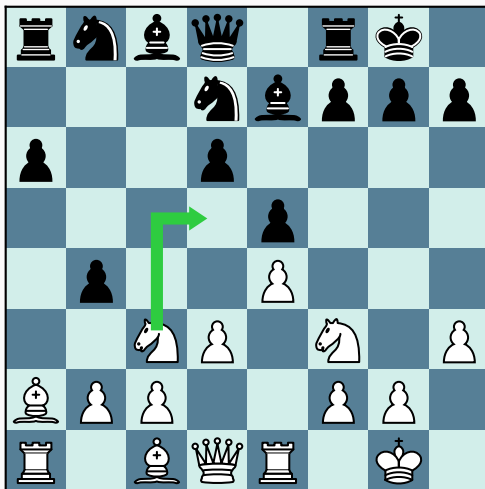
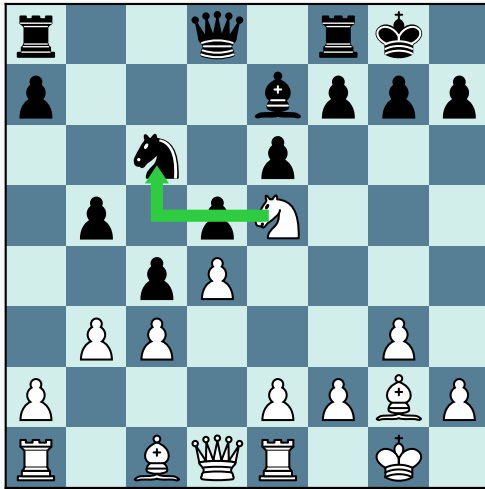
- increasing number of channels:
- $12 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 32$

ResNet

- 20 blocks with 64 channels



Strategy vs Tactics



Tactics

- Short term moves
 - winning a piece
 - checkmate
 - etc.

Strategy

- Long term move:
 - Good piece positioning
 - Pressure on opponent King
 - Control of center
 - etc.

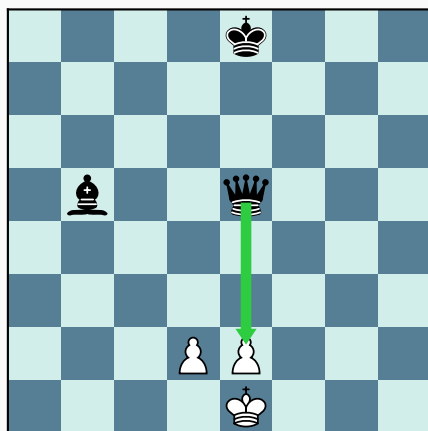
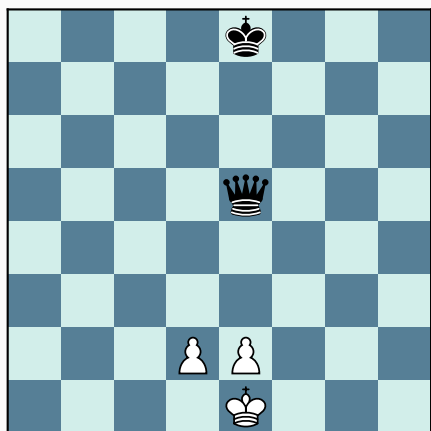
Imitation Learning: Behavior Cloning

Lichess Puzzles

- 6 million positions
- tactical positions

Bias

- no strategic positions
- no positions without tactic

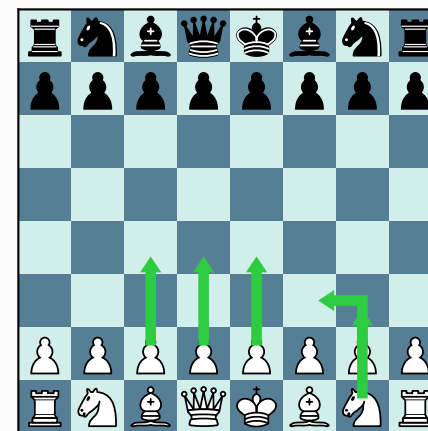
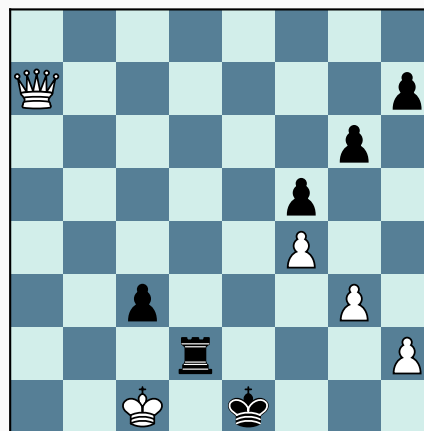


GM games

- chess.com master games
- balance tactics/strategy

Bias & Problems

- high level play
- many resignations
- multiple moves possible



Reward sets

0. **Sparse:** rewards only for true goal
 - reward for win/loss/draw
1. **Medium:** more rewards, but might misalign with goals
 - control center, material win, king safety
2. **Dense:** highest risk of misalignment
 - control outer center, control center, castling, promoting, blunder prevention, material, king safety, give check, win

Training

- Architecture: {FCNN, CNN, ResNet}
- Pretraining: {Puzzles, Masters} $\times$ {1 Epoch, 10 Epochs}, {Untrained}
- Reward Set: {R_0, R_1, R_2}
 - 1000 batches with 16 games each (REINFORCE)

→ 57 models

Evaluation

- All models plays 1000 games against all other model (same architecture)
- Normalized tournament score (win 1 point, draw 0.5 points, loss 0 points)

Results

Overall Results

arch rl	cnn				fc				resnet			
	none	r_0	r_1	r_2	none	r_0	r_1	r_2	none	r_0	r_1	r_2
train												
none	nan	0.266	0.235	0.251	nan	0.289	0.307	0.335	nan	0.408	0.407	0.399
pz_1	0.400	0.478	0.564	0.433	0.507	0.697	0.631	0.623	0.830	0.410	0.410	0.409
pz_10	0.446	0.672	0.515	0.520	0.530	0.637	0.521	0.558	0.925	0.403	0.403	0.403
gm_1	0.396	0.605	0.650	0.572	0.390	0.483	0.523	0.494	0.797	0.411	0.409	0.424
gm_10	0.434	0.714	0.694	0.654	0.418	0.494	0.538	0.526	0.870	0.387	0.398	0.399

- No optimal training strategy visible
- Sparse / No rewards seem to perform better

Let's take a closer look!

Behavior Cloning: Training Accuracy

dataset	epochs	fc	cnn	resnet
puzzle	1	62.0%	69.9%	74.6%
	10	65.8%	73.3%	81.7%
grand masters	1	31.0%	38.5%	41.5%
	10	31.8%	39.6%	43.7%

- Puzzle better generalizing
- **But** master games: different labels, same position

Behavior Cloning

BC models without RL

dataset	epochs	fc	cnn	resnet
puzzle	1	0.507	0.400	0.830
	10	0.530	0.446	0.925
grand masters	1	0.390	0.396	0.797
	10	0.418	0.434	0.870

BC model with RL

dataset	epochs	fc	cnn	resnet
none		0.310	0.251	0.405
puzzle	1	0.615	0.469	0.515
	10	0.562	0.538	0.533
grand masters	1	0.472	0.556	0.510
	10	0.494	0.624	0.513

- **Without RL:** Slightly better with 10 epochs of puzzle
- **With RL:** No pattern between pretraining and performance

Reward Shaping

RL models with BC

rewards	fc	cnn	resnet
none	0.461	0.419	0.855
r_0	0.520	0.547	0.404
r_1	0.504	0.532	0.405
r_2	0.507	0.486	0.407

RL models without BC

rewards	fc	cnn	resnet
r_0	0.289	0.266	0.408
r_1	0.307	0.235	0.408
r_2	0.335	0.251	0.399

- **Without Pretraining:** sparse / no rewards perform better than dense
- **With Pretraining:** No pattern between rewards and performance

Conclusion

- We could not find any training strategy across architectures
- Overall denser rewards hurt model performance

Possible Improvements

- **Data Prep:** Unify data in Masters games
- **Architecture:** Hyperparameter tuning on pretrain accuracy
- **BC:** Mix of master games and Puzzles
- **RL:** Longer training
- **RL:** Iterative reward shaping

How does it really play?

Well.... <https://lichess.org/cuRJkeJ1#82>